

2 Marks Questions

What is activated sludge? (2)

The activated sludge process is a type of wastewater treatment process for treating sewage or industrial wastewaters using aeration and a biological floc composed of bacteria and protozoa.

Mechanism of wastewater? (2)

Wastewater treatment is the process of converting wastewater – water that is no longer needed or is no longer suitable for use – into biologically clean water that can be discharged back into the environment. It's formed by a number of activities including bathing, washing, using the toilet, and rainwater runoff.

Diagnosis of mercury intoxication? (2)

A diagnosis we put when we have special clinical picture and information of anamnesis, which specify on a contact with mercury organic connections. The important diagnostic sign of intoxication is a presence of mercury in blood, urine, and at heavy intoxications – in a cerebrospinal liquid.

Thermophiles and its uses in PCR? (2)

Thermophiles are organisms that grow and thrive at temperatures (60–80°C) that are often too high for mesophiles—most thermophiles are Archaea. *Thermus aquaticus* is a thermophilic microorganism from which polymerase has been isolated and used for PCR (Taq-polymerase). DNA polymerases from other thermophiles are also being used in PCR.

Taq polymerase use in PCR?

Taq polymerase is a thermostable DNA **polymerase** I named after the thermophilic bacterium *Thermus aquaticus* from which it was originally isolated by Chien et al. in 1976. ... It is frequently **used** in the **polymerase** chain reaction (**PCR**), a method for greatly amplifying the quantity of short segments of DNA.

Extremophiles and their categories in nature?

A variety of microbes inhabit extreme environments.

Categories:

1. Psychrophiles
2. Thermophiles
3. Hyperthermophiles
4. Halophiles
5. Xerophiles
6. Alkaliphiles
7. Acidophiles

Name enzymes, solution and equipment's used in genetic engineering (2)

Enzymes:

Restriction endonuclease or restriction **enzymes**, DNA ligase, DNA polymerases and reverse transcriptase

Equipment's:

Gel Electrophoresis, plasmids, transformation or transduction, PCR, Identifying transgenic organisms

3 Marks Questions

What is wastewater and enlist its level?

Wastewater is "used water from any combination of domestic, industrial, commercial or agricultural activities, surface runoff or storm water, and any sewer inflow or sewer infiltration".

Wastewater treatment levels

Primary treatment

Primary treatment includes screens, sedimentation and grit removal.

Secondary treatment

Secondary treatment process removes carbonaceous organic matter and nutrients (nitrogen and phosphorus) from wastewater

Tertiary treatment

tertiary treatment includes filtering, disinfecting and preparing wastewater for recycling.

What is antibiotic and what is antibiotic resistance? 3

An **antibiotic** is a type of antimicrobial substance active against bacteria and is the most important type of antibacterial agent for fighting bacterial infections.

Antimicrobial resistance (AMR or AR) is the ability of a microbe to resist the effects of medication that once could successfully treat the microbe. The term **antibiotic resistance (AR or ABR)** is a subset of AMR, as it applies only to bacteria becoming resistant to antibiotics . **(Internet)**

What is microbial pesticides name microbial biopesticides?

Pesticides based on microorganisms and their products have proven to be highly effective, species specific and eco-friendly in nature. leading to their adoption in pest management strategies around the world.

Name: Biofungicides, bioherbicides, bioinsecticides **(Internet)**

What is the role of hormones in body and young children?

Hormones are the body's little messengers, which get produced in one part of the body, such as the thyroid, adrenal or pituitary gland, pass into the bloodstream or other body fluid and go to distant organs and tissues where they act to modify structures and functions.

Thermophiles and application of TAQ polymerase in PCR?

Thermophiles:

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Wastewater treatment definition?

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How microbes degrade polycyclic aromatic hydrocarbons?

Various **bacteria** have been found to **degrade** PAHs, in which **degradation** of naphthalene and phenanthrene has been most widely studied. ... Principally, **bacteria** favor aerobic conditions for **degradation** of PAHs via oxygenase-mediated metabolism

Bacteria manipulation without genetic engineering in biotechnology?

A general procedure is to take a sample of bacteria from, at, or near, the site of contamination from which a pure culture is obtained in the laboratory and identified, using standard microbiology techniques. Improve the bacterium's tolerance to the pollutant or to increase the capabilities of its pathways. Tolerance may be increased by culturing in growth medium containing increasing concentrations of the pollutant.

5 Marks Questions

Recombinant microorganisms, yeast and bacteria 5 marks

Recombinant bacteria is a type of bacteria which have the genetic material DNA of the another organism. Recombinant DNA is a technology which is developed by scientists that made it possible to insert a gene into the genetic material of a bacterium. This “recombinant” micro-organism can now produce such type of protein that coded by the gene. The scientists made the human insulin gene in the laboratory by using this technology.

Uses

The Recombinant bacteria is used in different ways to produce different products which are given below:

- Recombinant chymosin
- Recombinant human insulin
- Recombinant human growth hormone

Recombinant yeast:

The yeast in which introduce the characteristics according to our desire is called the recombinant yeast. Yeasts are unicellular eukaryotic microbial host cells that offer unique advantages in producing biopharmaceutical proteins.

Detail note on environmental biotechnology?

- Environmental biotechnology is the branch of biotechnology that addresses environmental problems, such as the removal of pollution, renewable energy generation or biomass production, by exploiting biological processes.
- Environmental Biotechnology deals with apparently less dramatic topics, though their importance may be every bit as great, and thus their direct relevance is far less appreciated by bulk of the population.
- The goals of Environmental Biotechnology are in everybody's interest... But for most it is simply addressing a problem that could have avoided already.

What are dioxins and write their sources?

Dioxins are a group of chemically-related compounds that are persistent environmental pollutants (POPs). Dioxins are found throughout the world in the environment and they accumulate in the food chain, mainly in the fatty tissue of animals.

Sources:

Dioxins are mainly by-products of industrial processes but can also result from natural processes, such as volcanic eruptions and forest fires. Dioxins are unwanted by-products of a wide range of manufacturing processes including smelting, chlorine bleaching of paper pulp and the manufacturing of some herbicides and pesticides.

What is surfactants and what is its purpose?

Surface-active agents /surfactants molecules which are active at interfaces and possess both polar(hydrophilic) and nonpolar(hydrophobic) characteristics in the same molecule.

Surfactants

lower the surface tension (or interfacial tension) between two liquids or between a liquid and a solid.

When surfactants are dissolved in water they orientate at the surface so that the hydrophobic regions are removed from the aqueous environment.

Purpose:

Surfactant reduces the surface tension of fluid in the lungs and helps make the small air sacs in the lungs (alveoli) more stable. This keeps them from collapsing when an individual exhales. In preparation for breathing air, fetuses begin making surfactant while still in the womb.

Write a note on zooplankton? 5

Zooplankton are heterotrophic plankton. Plankton are organisms drifting in oceans, seas, and bodies of fresh water. The word "zooplankton" is derived from the Greek zoon, meaning "animal", and planktos, meaning "wanderer" or "drifter".

Ecology:

Zooplankton is a categorization spanning a range of organism sizes including small protozoans and large metazoans. It includes holoplanktonic organisms whose complete life cycle lies within the plankton, as well as meroplanktonic organisms that spend part of their lives in the plankton before graduating to either the nekton or a sessile, benthic existence.

Nutritional modes in zooplankton

Herbivores: feed primarily on phytoplankton

Carnivores: feed primarily on other zooplankton (animals)

Detritivores: feed primarily on dead organic matter (detritus)

Omnivores: feed on mixed diet of plants and animals and detritus

10 Marks Questions**What is symbiosis and symbiotic nitrogen fixation? 10****Symbiosis:**

interaction between two different organisms living in close physical association, typically to the advantage of both. Nitrogen fixation is a process by which nitrogen in the air is converted into ammonia (NH₃) or related nitrogenous compounds. Atmospheric nitrogen, which is molecular dinitrogen (N₂), is relatively nonreactive and is metabolically useless to all but a few microorganisms. The fixation process converts N₂ into ammonia, which is metabolized by most organisms.

Nitrogen fixation is essential to life because fixed inorganic nitrogen compounds are required for the biosynthesis of all nitrogen-containing compounds. As part of the nitrogen cycle, it is essential for agriculture and the manufacture of fertilizer. It is also, indirectly, relevant to the manufacture of all chemical compounds that contain nitrogen, which includes explosives, most pharmaceuticals, and dyes. Symbiotic nitrogen fixation occurs in plants that harbor nitrogen-fixing bacteria within their tissues. The best-studied example is the association between legumes and bacteria in the genus *Rhizobium*. Each of these is able to survive independently (soil nitrates must then be available to the legume), but life together is clearly beneficial to both. Only together can nitrogen fixation take place. A symbiotic relationship in which both partners benefit is called mutualism.

What is cloning vector? Why it is important in genetic engineering? (10)

A cloning vector is frequently a plasmid or a bacteriophage (bacterial virus). Must be fairly small and fully sequenced, able to replicate itself when reintroduced into a host cell, thus producing large amounts of the recombinant DNA for further manipulation. Also it must carry on it 'selector marker' genes, which are indicators of genomic integrity and activity. A common design of a cloning vector is one which carries two genes coding for antibiotic resistance. The 'foreign' gene is inserted within one of the genes so that it is no longer functional; therefore it is possible to discriminate by standard microbiology techniques, which bacteria are carrying plasmids containing recombinant DNA and which are not. To make the manipulations easier, selector marker genes normally contain a multi cloning site (MCS) which is a cluster of sites for restriction enzymes. Insertion of foreign DNA in vector at MCS site causes disruption of these marker genes resulting in loss of the function of that gene. For example, if that gene codes for antibiotic resistance, it will no longer be functional and will not protect the bacterium from that antibiotic.

Viruses may be used as vectors i.e., the DNA of viruses can be used to carry foreign DNA. Most such recombinant viral DNA vectors need phage particles to perform proper gene transfer i.e. transduction. Transfer of naked viral recombinant vector to host cells is called transfection, and tends to have lower uptake rates as compared to transduction.

Explain in detail the principles of transformation of the plants? 10

transformation is the genetic alteration of a cell resulting from the direct uptake and incorporation of exogenous genetic material from its surroundings through the cell membrane.

Principle:

There are number of principles which are used for the transformation of the plants which are given below:

Vector method:

There are two vector methods which are given below:

Agrobacterium

Viral transformation

Vector Free method:

There are different vector free methods which are given below:

Gene gun

Electroporation

Vector method

There are two vector methods which are given below:

Agrobacterium

Viral transformation

Agrobacterium:

Agrobacterium mediated transformation is the easiest and most simple plant transformation. Plant tissue (often leaves) are cut into small pieces, e.g. 10x10mm, and soaked for 10 minutes in a fluid containing suspended Agrobacterium. The bacteria will attach to many of the plant cells exposed by the cut. The plant cells secrete wound-related phenolic compounds which in turn act to up regulate the virulence operon of the Agrobacterium.

Vector method

Viral Transformation:

Package the desired genetic material into a suitable plant virus and allow this modified virus to infect the plant. If the genetic material is DNA, it can recombine with the chromosomes to produce transformant cells. However, genomes of most plant viruses consist of single stranded RNA which replicates in the cytoplasm of infected cell. For such genomes this method is a form of transfection and not a real transformation, since the inserted genes never reach the nucleus of the cell and do not integrate into the host genome. The progeny of the infected plants is virus-free and also free of the inserted gene.

Vector less method

There are two vector less method which are given below:

Gene gun

Electroporation

Gene Gun:

It is referred as particle bombardment or biolistics. Particles of gold or tungsten are coated with the DNA and then shot into young plant cells.

Electroporation:

Formation of transient holes in cell membranes using electric pulses of high field strength this allows DNA to enter as described above for bacteria.

Marks Not Confirm

What is Recombinant?

Recombinant is the anything which have the mixture of two hereditary materials one is of the organism and the second is of the another which hereditary material is introduced into the body of first.

Advantages of Yeast in Recombinant?

Yeasts have been used for thousands of years in food and fermentation processes to produce alcoholic beverages and breads. In recent times, the industrial importance of yeasts has extended beyond its traditional use in fermentation into various healthcare sectors, such as in the production of therapeutic recombinant proteins. Yeast cells are particularly advantageous as hosts for biopharmaceutical production in that they are generally recognized as safe (GRAS) organisms.

Types of pollution?

There are different types of the pollution which are given below:

Air pollution

Water pollution

Land pollution

Environmental pollution

Light pollution

Noise pollution

Mitochondrial disease & diabetes mellitus

Mitochondrial diseases are long-term, genetic, often inherited **disorders** that occur when **mitochondria** fail to produce enough energy for the body to function properly.

Diabetes mellitus (DM), commonly known as **diabetes**, is a group of metabolic disorders characterized by high blood sugar levels over a prolonged period. Symptoms of high blood sugar include frequent urination, increased thirst, and increased hunger. **(Internet)**

Transformation Plant?

Plant transformation is a scientific approach whereby DNA from any organism is inserted into the genome of a species of interest. The inserted DNA is called a “transgene”, and the resulting plant is said to be “transgenic”. Transgenic plants are vital for both research and agriculture.

Bioaccumulation?

The collection of substances e.g. toxins and chemicals, in an organism's body is called bioaccumulation.

Sources: substances from any source e.g. air, water, soil, could be stocked through bioaccumulation.

Substances to be gathered through bioaccumulation have following properties;

They are preferably relentless, lipophilic and stable

Unable to translocate throughout the body

Have low catabolic rate

High affinity for proteins

Pollutants take entry to the organism's body or tissue through inhalation, ingestion, epithelial contact or by additional means.

What is toxicity?

Toxicity is the degree to which a chemical substance or a particular mixture of substances can damage an organism. Toxicity can refer to the effect on a whole organism, such as an animal, bacterium, or plant, as well as the effect on a substructure of the organism, such as a cell (cytotoxicity) or an organ such as the liver (hepatotoxicity).

Use of recombinant yeast?

Yeast in medicine Industries

Pharmaceutical proteins comprise one of the fastest growing groups of molecular medicines and currently play important

roles in the treatment of many diseases. The estimated market value of biopharmaceuticals, including recombinant therapeutic proteins, nucleic acid-based products, and engineered cell-based products.

What is plac z expression vector?

Lac –Z promoter operator is in frame with lac-Z alpha fragment (the NH₃ terminal part of Galactosidase gene.

Multiple cloning sites are found in the border of NH₃ end including ATG sequence.

The presence of such restriction site sequences should not disturb the functional activity of the protein, which complements with the omega fragment of the Lac-Z produced by the bacterial cell as the complement.

If any gene is placed in proper frame in the MCS the protein expressed will be in fused form.

The expression of the gene can be regulated by IPTG (Isopropyl thio β-Galactoside).